✅ 1. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

- Highest relevance.
- The apartment floor plan is an irregular polygon (with balconies, wall indentations, etc.). Calculating room-by-room would be tedious.
- The prompt hints at a "more efficient method," encouraging students to move beyond local thinking and restructure the entire shape.
- The optimal strategy is to treat the whole apartment (including the balcony) as a large rectangle minus one or more smaller rectangles (e.g., an entrance recess or internal cutout), or to recognize its outer boundary as a composite rectangle definable by just four dimensions.
- For example, only the following might be needed:
  - Total width (W)
  - Total height (H)
  - Additional width/height of a protruding balcony
  - Or dimensions of an internal "notch" to subtract
- This process exemplifies geometric restructuring to reduce a complex region to combinations of basic shapes, drastically minimizing measurement effort—precisely what Competency #7 describes.

✅ 2. Competency #5: Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

- Highly relevant.
- Students must observe the geometric structure of the floor plan and recognize that it is composed of rectangles joined or subtracted—a key pattern.
- Once they see that "the entire shape can be enclosed by a bounding rectangle with minor corrections," they understand why only four lengths suffice.
- This recognition of a rectilinear polygon structure (made of axis-aligned rectangles) is essential to finding the efficient solution.
- If they only see "a collection of rooms," they miss the big picture; if they see "one large block with a protrusion," they grasp the structural essence.

✅ 3. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying significant variables

- Highly relevant.
- Students must extract from the diagram the dimensions that actually determine total area, ignoring irrelevant details.
- For instance, in an L-shaped apartment, total area = area of large rectangle − area of missing corner. Only four measurements are needed:
  - Outer length and width
  - Length and width of the missing section
- This requires distinguishing essential variables from redundant information (e.g., interior partition walls) and focusing on the degrees of freedom that control total area.
- This is core to modeling: capturing system essence with minimal parameters.

✅ 4. Competency #3: Select appropriate mathematical representations to capture essential features of a problem

- Moderately relevant.
- The problem provides one representation (a scaled floor plan), but students must mentally or graphically construct a more effective one: a bounding rectangle plus correction terms.
- By marking four key line segments on the diagram, students are effectively creating a visual representation of a simplified model.
- This shift from a "detailed architectural drawing" to an "area-calculation model" constitutes a representation transformation, aligning with Competency #3.

🟡 5. Competency #12: Identify the constraints, assumptions, and simplifications in a mathematical model

- Weak but present relevance.
- The efficient method relies on implicit assumptions:
  - All walls are vertical/horizontal (orthogonal structure)
  - The balcony lies in the same plane as the main unit
  - Minor curves or angled corners are ignored (the diagram uses only straight lines)
- If the apartment had curved walls or diagonals, the "four-length" method would fail. A student who recognizes that this approach depends on the simplifying assumption of "rectangular composition" engages with this competency.

✅ 1. 能力 #7：简化或分解情境，使其便于数学分析

> *"Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis"*

- 最高相关性。
- 公寓平面图是一个不规则多边形（含阳台、墙体等凸凹结构）。直接分房间计算繁琐。
- 题目提示存在“更高效的方法”，引导学生跳出局部思维，重构整体形状。
- 最优策略是：将整个公寓（含阳台）视为一个大矩形减去一个或多个小矩形（例如入口凹槽或内嵌区域），或者直接识别其外轮廓为一个可由4条边定义的复合矩形。
- 例如，只需测量：
  - 总宽度（W）
  - 总高度（H）
  - 若有突出阳台，则测量阳台的额外宽/高
  - 或测量内部“缺口”的尺寸以做减法
- 这一过程正是通过几何重构将复杂区域简化为基本图形组合，极大减少测量次数——完美体现能力 #7。

✅ 2. 能力 #5：识别问题中的数学结构（如规律、关系、模式）

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- 高度相关。
- 学生需观察平面图的几何结构，识别其由矩形拼接或切割而成这一关键模式。
- 一旦意识到“整个图形可被外接矩形覆盖，仅需修正少数部分”，就能理解为何只需4个长度。
- 这种对空间组合结构（rectilinear polygon）的识别，是找到高效解法的前提。
- 若只看到“一堆房间”，就无法发现整体性；若看到“一个带凸起的大方块”，就能抓住结构本质。

✅ 3. 能力 #6：识别并描述现实问题中的数学要素，包括关键变量

> *"Identify and describe the mathematical aspects... including identifying the significant variables"*

- 高度相关。
- 学生必须从图纸中提取决定总面积的关键维度，而非所有细节。
- 例如，在 L 形公寓中，总面积 = 大矩形面积 − 缺口矩形面积，所需变量仅为：
  - 外部长、宽
  - 缺口长、宽（共4个）
- 这要求学生区分必要变量与冗余信息（如单个房间隔墙），聚焦于控制整体面积的自由度。
- 这正是建模中“变量选择”的核心：用最少参数捕捉系统本质。

✅ 4. 能力 #3：选择合适的数学表示方式来刻画问题

> *"Select appropriate mathematical representations to capture essential features of a problem"*

- 中等相关。
- 题目本身提供了一种表示（平面图 + 比例尺），但学生需在脑中或图上构建另一种更有效的表示：整体边界框 + 修正项。
- 通过在图上标记4条关键线段，学生实际上是在创建一个简化模型的可视化表示。
- 这体现了从“详细建筑图”到“面积计算模型”的表示转换，属于能力 #3 的范畴。

🟡 5. 能力 #12：识别模型中的约束、假设与简化

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- 弱相关但存在。
- 高效方法隐含假设：
  - 所有墙体垂直/水平（正交结构）
  - 阳台与主体在同一平面
  - 忽略微小转角或弧形（题目图示为纯直线）
- 若公寓含曲线或斜墙，则4长度法失效。学生若能意识到此方法依赖于“矩形组合”这一简化假设，则触及此项能力。